Max Simmonds

Senior Backend Engineer

Contact

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Achievements

- Engineering Act of the Week - Starship, 2026
- Winner - Build Bolt Week Hackathon, 2024
- First Class Master's - Average 84%
- Published paper in IEEE
- Presented at the Royal Institute of Science
- Voted Bright Spark, 2019
- Manufacturer Top 100, 2015
- Winner of NASA Hackathon, 2013

Biography

Backend engineer with close to a decade of experience across robotics, fintech, banking, and aerospace. Currently at Starship Technologies in Tallinn, owning the energy, charging, battery, and tampering-detection domains end-to-end for the autonomous-delivery fleet - designing, modelling, and shipping production services in Go, with the data-science work to back them. Previously at Bolt (Rides/Marketplace/Pricing) and Lloyds Banking Group (5000 TPS Go microservices on OCP).

A hardware-to-software background (CERN, ESA, Open Cosmos, Thales) means I'm equally comfortable on the systems and on the maths.

Relevant Experience

Starship | Backend Software Engineer **Feb 2025 - Present**

Core Services

Tallinn, EE

- **Battery modelling, end-to-end.** Owned the per-service-area battery-range regression model end-to-end: exploratory analysis and back-testing in Databricks notebooks, feature engineering across a year of delivery, weather, and platform telemetry from the global fleet, training, bias analysis, monitoring. **Reduced prediction error from $\sim$100% MAPE to $\sim$25% MAPE ($\sim$4× more accurate)**, increasing robot utilisation and reducing in-field depletion. Drove global rollout with weekly monitoring and a self-righting retraining CronJob; awarded **Engineering Act of the Week**.
- **Tampering detection.** Designed and built a new microservice from scratch (PostgreSQL schema with migrations, concurrent MQTT consumers, Kafka producers). Modelled detection as the **dot product of at-rest vs live gravity vectors**, with per-axis noise-floor analysis to suppress wind, pedestrians, and curb-knocks. Ingests live MQTT telemetry from the production fleet; triggers automatic robot wake-up via a Kafka alert consumer; runs fleet-wide.
- **Re-architected fleet diagnostics** from synchronous request/response to event-driven Kafka with per-command topics ($\sim$15k lines across the feature). **Now the default production path company-wide.**
- **Re-introduced the company's fast-charger stack** end-to-end: backend service, React/TypeScript admin-panel UI, Kafka topics for connector state, battery-management-slot lifecycle, wireless-charging detection. Migrated a core internal service from MongoDB to PostgreSQL and shipped new capability- and version-tracking APIs.
- **Drove supply-chain hardening org-wide**: 12+ Go services migrated to SHA-pinned distroless Docker Hardened Images, secret-leak closed via image history, 14-day dependency cooldown adopted across 8+ repos, Trivy scanning added to shared CI.

Skills

Languages
- Golang
- C
- Python
- JavaScript/Typescript
- SQL

Backend APIs
- gRPC/Protobuf
- REST
- Kafka
- MQTT
- Event-driven architecture
- System design/Design docs

Data & ML
- PostgreSQL/MongoDB
- Databricks/dbt
- pandas/Numpy/SciPy
- Regression Modelling
- Backtesting/feature engineering

Infra & DevOps
- Kubernetes/Kustomize
- Docker
- GCP/AWS
- Terraform
- Jenkins/GitHub Actions
- Linux/Unix/ROS

Bolt | Backend Software Engineer — **Jul 2024 - Jan 2025**
Rides, Marketplace, Pricing team
Tallinn, EE

- Designed offline driver notification microservice from scratch, targeting every single driver globally. From initial concept, to design doc, through to end product and A/B testing.
- Winner of Build Bolt Week Hackathon, designing a production ready feature in 4 days. I implemented data aggregation, data transformation and parsing. I presented the final product on stage to the company.
- Implemented fraud prevention feature, used by millions, saving an estimated 120,000 EUR per year
- On-duty rotations: identified root causes quickly and communicated solutions clearly across teams. Contributed React/TypeScript frontend features to Bolt's admin panel alongside backend work.

Lloyds Banking Group | Senior Software Engineer — **Jan 2023 - Jul 2024**
Chief Technical Office, Center of Excellence
London, UK

- Transformed a high-load (5000 TPS) Java Spring Boot API into a Golang microservice using Kafka and IBM DB2, deployed to OCP.
- **Designed the bank's production-ready Go microservice template** - gRPC + gRPC gateway, in-memory testing DB, protobuf-driven OpenAPI/Swagger - adopted across the Centre of Excellence.
- Presented to audiences of up to 450 engineers; ran workshops on the Go template.

TEO Robotics Ltd | CEO & Founder — **Apr 2021 - Feb 2025**
Electronic Design Consultancy
Bristol, England

- Founded an electronic-design consultancy in Bristol from no initial investment; customers across hydrogen fuel cells, medical devices, mining, and music tech.
- Python codegen, FPGA / Verilog at speeds up to 800 MHz, firmware in C for embedded systems.

Thales Alenia Space | Technical Lead — **Oct 2021 - Jan 2022**
Cryogenic Coolers
Bristol, UK

- Wrote firmware and register level drivers / algorithms in C (UART, CLI terminal, numerically controlled oscillator) streamlined testing (3x faster)
- Wrote MATLAB scripts to run system identification (mathematical models) of captured data, generating electrical equivalent models, and bode plots
- Work package manager, hour tracking/estimating, communicating with manufactures and stake holders. Managed several teams

Diamond Light Source | Power Supplies Engineer — **Sept 2020 - Oct 2021**
UK's National Synchrotron
Didcot, England

- Reduced test-data ingest by several orders of magnitude; introduced automatic alerts for anomalies; built a MySQL + Grafana stack on Kubernetes.
- Designed ultra-sensitive (0.000001 A) current sensors and new power supplies for the UK's national synchrotron.

Selected Projects

Drone Hacking (2024)
Reverse-engineered the Snaptain AF15 protocol to fly it from a laptop: packet capture, RF analysis, custom Python client.

Numerically Controlled Oscillator
32/64-bit fixed-point complex-phasor synthesis in C on STM32 driving a cryogenic cooler at 100 kHz PWM.

Publications

- "Discrete-time modelling of pulse-width modulated DC-DC converters in sub-sampling conditions". In: *2018 IEEE 19th Workshop on Control and Modelling for Power Electronics (COMPEL)*, 25-28th Jun, 2018.
- "Fibre Optic Notch Filter For The Antiproton Decelerator Stochastic Cooling System". In: *CERN Document Server*, 24th Aug, 2016.

Certificates

- Russian Language (60+ hrs) - Preply, 2023
- Project Management - Safran, 2019
- NVQ Level 2 - Engineering

Interests

- Rock Climbing
- Motorbikes
- Engineering
- Coding
- DIY

Open Cosmos | Electronics Engineer — **Jul 2019 - Sept 2020**
New Space Start up
Didcot, England

- Wrote Python scripts for automating tests and control of space hardware. Reduced time to perform measurements by several days, significantly reduced risk to flight hardware
- Used PyFTDI for the hardware-software interface.

Safran (UK) Electrical & Power | Senior Specialist Engineer — **Aug 2018 - Jul 2019**
Research and Technology Division
Pitstone, England

- Captured and managed FPGA firmware design requirements to firmware team, working closely with them to ensure correct code
- Role mainly focused on electronics, digitising the Airbus A380 Generator Control Unit (GCU)

European Space Agency (ESA) | Power Systems Engineer — **Jul 2017 - Jul 2018**
Power Conditioning and Distribution
Noordwyk, Netherlands

- Programmed and designed a digital controller for a high frequency switch mode converter, in VHDL, allowing for the publication of a paper in IEEE
- Implemented a fixed point PID controller with ADC reader
- Co-authored a paper on my research, assessing the feasibility of digitally controlled high frequency switch mode converters for spacecraft

CERN — **July 2016 - August 2016**
Application Engineering Summer Student
Geneva, Switzerland

- Wrote LabVIEW drivers for controlling state-of-the-art optical equipment and thermoelectric controllers. Alleviating a project on hold for several years and reducing it's size by an order of magnitude
- Reverse engineered, upgraded, & restored existing fibre optic notch filter, used in the Antiproton Decelerator at CERN

National Instruments (NI) — **July 2014 - August 2015**
Application & Technical Marketing Engineer
Newbury, England

- One of two interns (out of 16) to pass 'Certified LabVIEW Developer'; developer of the world-first myRIO-powered musical Tesla coil (youtu.be/AyXX_V5bcWM).

Education

MEng Electrical & Electronic Engineering — **2012 - 2017**
University of Plymouth

- First class degree, with honours. Master's thesis: Inertial Navigation System with mathematical modelling (Earth's rotation across reference frames); built an electric self-balancing unicycle as a side project.

29th May 2026

Max Simmonds